

Executive Summary - Unit Economics: CAC, LTV & Cross-Sell Propensity

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The Headline

Paid search is a stronger acquisition channel than organic search once you look past lead volume to activated revenue. Over half of "won" sellers never sell a single item. Customer lifetime value is almost entirely a **first-purchase** number, since 97.0% of customers never return. And freight cost on its own explains why electronics and remote-state orders come in far below the platform's average margin.

Four Findings, Each With a Number

- 1. Channel quality $\neq$ channel volume.** Organic search brings in the most leads (2,296 MQLs) but paid search converts more efficiently once activation and realised GMV are counted (channel value score 0.71 vs 0.63). Referral leads are rare (284) but each is worth roughly 2-3x the median activated seller (R\$1,368 vs R\$512-640).
- 2. Winning a seller is not the same as activating one.** 842 of 8,000 MQLs became sellers (10.5% win rate), but only 379 of those 842 (45.0%) ever sold anything in the observed window. Any channel-ROI conversation that stops at "win rate" is measuring the wrong thing.
- 3. LTV growth almost stops after month 0.** Cohort cumulative margin per customer moves from R\$111.41 at first purchase to R\$123.45 by month 12 - a 10.8% lift over a full year, produced entirely by the 3.0% of customers who buy again. That is what a marketplace checkout with no loyalty program looks like, and it points at retention as the lever, not lifetime-value expansion among existing repeat buyers.
- 4. Freight, not price, decides who is profitable.** Electronics is a high-volume category (2,755 items, R\$157k GMV) at only 31.5% margin because freight eats 68.6% of price - vs. 83.0% margin on watches_gifts, a comparably-sized category. The same pattern repeats geographically: São Paulo (where most sellers are based) runs 73.5% margin; Amazon-region states (RO, RR) run ~40%, purely from distance-driven freight cost.

What This Doesn't Claim

There is no dollar CAC or payback period here. Olist doesn't disclose marketing spend by channel, and no outside benchmark was substituted for it (see docs/seller_cac_methodology.md). No payment-processing fee or platform take-rate is netted into any margin figure (see docs/margin_proxy_definition.md).

Cross-Sell Propensity

A logistic model on an out-of-time split (train through Feb 2016, test Mar-May 2016) predicts next-product adoption with AUC 0.85-1.00 across 16 of 24 Santander products. The other 8 had too few positive cases to score meaningfully and were dropped. The output is a ranked top-3 next-best-product list per customer - see outputs/tables/cross_sell_ranked_list_sample.csv.